

Quiz: Instruction formats

This quiz has 5 questions. The passing grade is 4.

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Started on	Wednesday, January 15, 2025, 12:06 AM
State	Finished
Completed on	Wednesday, January 15, 2025, 12:11 AM
Time taken	4 mins 49 secs
Grade	2.00 out of 5.00 (40%)

Question **1**

Correct

1.00 points out of 1.00

In z/Architecture, the first two bytes of an instruction always contain the first operand's addressing halfword

Select one:

- ☐ True
- ☒ False ✓

The first byte of an instruction is always the Opcode, so can't be part of an operand's addressing halfword.

The correct answer is 'False'.

Question **2**

Incorrect

0.00 points out of 1.00

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If the second operand in an RX-type instruction is written as '10(7,2)', this means

Select one:

- ☐ A. Base register 7, index register 2, displacement 10
- ☐ B. Base register 2, index register 10, displacement 7
- ☐ C. Base register 2, index register 7, displacement 10
- ☐ D. Base register 10, index register 2, displacement 7
- ☒ E. Base register 10, index register 7, displacement 2 ✖
- ☐ F. Base register 7, index register 10, displacement 2

The Assembler syntax for the operand of an RX-type instruction is:

displacement(index register, base register)

So in '**10(7,2)**':

10 = displacement

7 = index register

2 = base register

The correct answer is: Base register 2, index register 7, displacement 10

Question **3**

Incorrect

0.00 points out of 1.00

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In an SS-format instruction

Select one:

- ☐ A. Both operands are registers
- ☐ B. One operand is a register and another a base-displacement storage address
- ☐ C. One operand is a register and another an indexed storage address
- ☐ D. One operand is a register and another a 16-bit relative address
- ☐ E. Both operands are storage addresses
- ☒ F. One operand is a register and another a long displacement storage address **✖**
- ☐ G. One operand is a storage address and another an immediate value

The correct answer is: Both operands are storage addresses

Question **4**

Incorrect

0.00 points out of 1.00

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In an RI-format instruction

Select one:

- ☐ A. One operand is a register and another a base-displacement storage address
- ☐ B. Both operands are storage addresses
- ☒ C. One operand is a register and another an indexed storage address **✖**
- ☐ D. One operand is a register and another a 16-bit relative address
- ☐ E. One operand is a register and another a long displacement storage address
- ☐ F. Both operands are registers
- ☐ G. One operand is a storage address and another an immediate value

The correct answer is: One operand is a register and another a 16-bit relative address

Question **5**

Correct

1.00 points out of 1.00

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The CPU picks the next instruction for execution from

Select one:

- ☒ A. The PSW Instruction Address (IA) field ✔
- ☐ B. The base register
- ☐ C. The Next Instruction Register (NIR)
- ☐ D. General register 0
- ☐ E. Memory address 0

Correct!

The CPU always picks the next instruction to execute from the PSW Instruction Address (IA).

The correct answer is: The PSW Instruction Address (IA) field